

Rudra Mehta

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EDUCATION

Dharmsinh Desai University

Bachelor of Technology in Information Technology | CPI: 8.61 / 10.0

Nadiad, Gujarat, India

Expected May 2027

TECHNICAL SKILLS

Languages: Java, C++, Python, SQL

Frameworks and Technologies: Spring Boot, React, Node.js, WebSockets, WebRTC, gRPC

Cloud and DevOps: AWS, Docker, Git

Databases: PostgreSQL, MySQL, MongoDB, Redis

Core Concepts: System Design, Microservices, Backend Architecture, REST APIs, DBMS

PROJECTS

Studify | *Real-Time Scalable Smart Classroom Platform* | 📁 Code

Java, Spring Boot, WebRTC, WebSockets, AWS, MySQL

- Architected a backend supporting **real-time meetings (50 concurrent users)** and **broadcast streaming (300+ viewers)** via WebRTC Selective Forwarding Unit architecture with event-driven participant synchronization.
- Deployed high-throughput microservices on **AWS ECS** processing **1,000+ API requests per minute** with JSON Web Token authentication and fine-grained role-based access control.
- Provisioned distributed cloud infrastructure using **Application Load Balancer**, private networking, and horizontal auto-scaling; tuned database queries to cut response latency by **40 percent**.

IrisPay | *AI-Powered Biometric Payment Gateway* | 📁 Code

Java, Spring Boot, Python, Flask, TensorFlow, PostgreSQL

- Built a **microservices-based** payment system coupling a transaction service with an ML inference service to enable secure, card-free biometric authentication at the point of payment.
- Engineered a **facial embedding pipeline** with DeepFace and TensorFlow using cosine similarity verification; persisted only **AES-encrypted embeddings**, removing raw biometric image storage entirely.
- Protected inter-service traffic via **JSON Web Token authentication** and an API gateway.

QuickFix | *On-Demand Home Services Platform* | 📁 Code

Java, Spring Boot, PostgreSQL, Docker, JWT, WebSockets

- Delivered scalable RESTful backend services supporting **100+ concurrent users** through connection pooling and query optimization, reducing average API response time by **35 percent**.
- Enforced stateless security using **JSON Web Tokens** and Spring Security with role-based access control, refresh token rotation, and full session lifecycle management.
- Implemented an event-driven live-update layer via **WebSockets** for real-time booking notifications; containerized the full stack with **Docker** for reproducible cross-environment deployments.

CodeIntel | *AI-Powered Codebase Intelligence Platform* | 📁 Code

Java, Python, gRPC, Redis, Spring Boot, FastAPI

- Designed a **microservices architecture** using Spring Boot for orchestration and Python FastAPI for Abstract Syntax Tree parsing; preserved class and method-level semantic boundaries through structured code chunking.
- Constructed a **retrieval-augmented generation pipeline** with hybrid retrieval combining BM25, keyword scoring, and dependency graph expansion for accurate multi-hop context resolution.
- Built a gRPC-based query orchestrator with **Redis caching** and SHA-256 request hashing; connected **Gemini API** to detect architectural issues and produce system-level engineering insights.

ACHIEVEMENTS AND CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02): Earned foundational certification in Amazon Web Services, covering cloud concepts, AWS core services, security, pricing models, billing, and architectural best practices.

DUHacks 5.0 Hackathon: IrisPay ranked among the **Top 10 offline projects** out of **200 participating teams** for innovation in AI-powered financial technology.

LeetCode: Solved **300+ algorithmic problems**